Legends: Monsters and Heroes

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It's a funny role-playing game written in Java. Lead a party of 1–3 heroes across a board world, fight scaled monsters on common ground, earn gold and experience, and spend gold in markets on weapons, armor, potions and spells. If every hero faints, the game is over.

From the project

- **data** is the folder with the .txt stat files
- Requires Java 8 or newer.
- Typing **Q/q** at any prompt quits the game immediately.

How to play

The map is re-rendered every turn:

P = your party M = market (blank) = common ground (will encounter a monster)

X = inaccessible show: `That way is blocked.`

Outside battle

Key	Action
W/A/S/D	Move up / left / down / right
I	Show hero information (level, HP, MP, XP, gold, skills, equipment)
E	Manage inventory: equip weapon/armor, drink potions
M	Enter the market (only while standing on an M tile)
H	Show the help text again
Q	Quit (works at any prompt)

In battle — each living hero acts once per round; type the number of the action:

#	Action
1	Attack with the equipped weapon
2	Cast a spell (costs MP; spells are single-use)
3	Use a potion
4	Equip a weapon or armor (uses the turn)
5	Show full stats of heroes and monsters (free, does not use the turn)

Project Structure

File	Description
data	Stores the default game data in plain text files.
Main.java	Entry point of the program. Initializes GameData and starts the game.
Game.java	Main game controller. Handles party setup and the exploration loop.
Rules.java	Centralizes all numerical game rules and constants, making game parameters easy to maintain and modify.
Console.java	Handles all console input and output, keeping user interaction separate from game logic.
Character/Entity.java	Abstract base class for all characters. Defines shared attributes such as name, level, and HP for Hero and Monster.
Character/Hero.java	Abstract base class for heroes. Manages skills, gold, experience, inventory, equipment, and leveling. Contains nested static classes: Warrior, Sorcerer, and Paladin.
Character/Monster.java	Abstract base class for monsters. Defines shared combat attributes such as damage, defense, and dodge. Contains nested Dragon, Exoskeleton, and Spirit classes, each with a favored stat.
Character/Item.java	Abstract base class for items. Defines shared attributes such as name, price, level, and durability. Contains nested classes Weapon, Armor, and Potion.
Character/Spell.java	Defines spells and the SpellEffect strategy. Contains nested Ice, Fire, and Lightning spell classes.
Character/Party.java	Manages the hero party and its current position on the game map.
Character/GameData.java	Reads all data files once and provides factory methods for creating heroes, monsters, and market stock. Creates fresh objects for each game.
Character/Menus.java	Handles inventory-related menus, including equipment

	management, potion usage, and the market menu.
Battle/Battle.java	Core combat engine. Manages battle logic and combat execution without directly printing output.
Battle/BattleObserver.java	Observer interface that defines methods for receiving and handling battle events.
Battle/BattleObserverAdapter.java	Provides empty default implementations of observer methods, allowing subclasses to override only the events they need.
Battle/BattleLogger.java	First battle observer. Displays battle events and messages to the player.
Battle/BattleStats.java	Second battle observer. Tracks battle events and generates a battle statistics summary.
Market/Market.java	Manages market stock and buying/selling rules. Returns error messages instead of directly printing them.
World/World.java	Handles world grid generation, map rendering, and movement boundaries.
World/Tile.java	Abstract base class for map tiles. Contains three tile types: CommonTile, MarketTile, and InaccessibleTile.

Design Highlights

- **Using design pattern:** observer pattern Battle reports everything what happens to its BattleObservers and don not have to prints itself. BattleLogger turns events into text for the player; BattleStats counts the same events and prints a post-battle summary. New observers (sound effects, replay logging, …) can be added without touching Battle.

  Using strategy pattern (spells). A Spell holds a SpellEffect (IceEffect, FireEffect , LightningEffect). The spell does not know or care which effect it has — it just calls effect.apply(target). New spell methods are new one-line effect classes.
- Using generic. Hero.getItems(Class<T>) returns only the potions, spells, weapons or armor of a hero with one method.
- Console is the only class that reads from System.in; Rules is the only place with game numbers (world size, tile percentages, damage factors, XP curves, durability) that can be modified by the customer's needs flexibly.

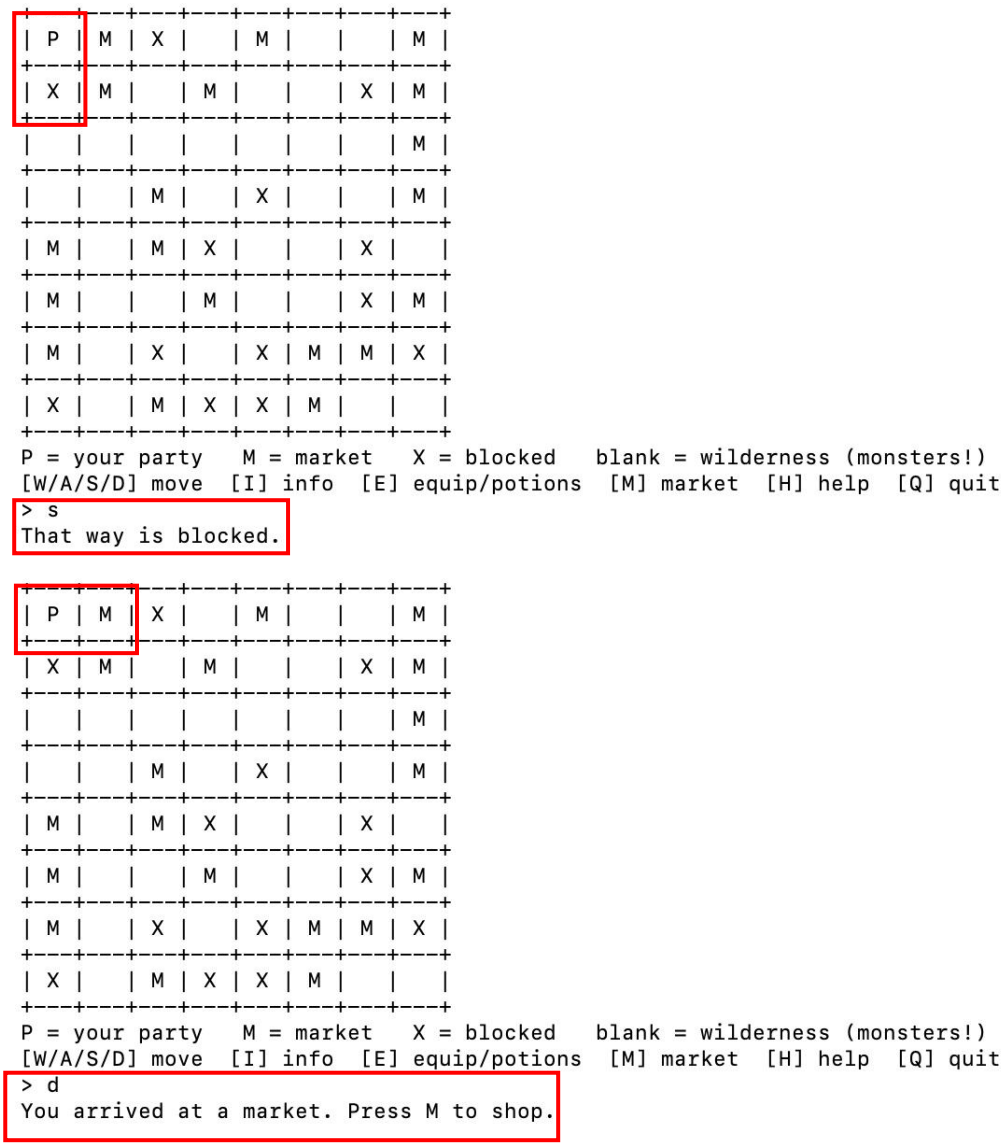
Output Screenshot

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==============================================
        LEGENDS: MONSTERS AND HEROES
==============================================
Lead a party of heroes across the land. Fight monsters to earn gold and
experience, then spend gold in markets on weapons, armor, potions and spells.

CONTROLS (outside battle):
  W/A/S/D  move up/left/down/right     M  enter market (when standing on M)
  I  show hero information             E  equip items / drink potions
  H  show this help again              Q  quit (works at any prompt)
In battle, type the number shown next to an action. Heroes win by defeating all
monsters; if every hero faints, the game is over. Walking on blank tiles may
start a battle. Fainted heroes are revived (at half HP/MP) after a victory.

How many heroes (1–3)? 2

Available heroes:
  1) Gaerdal Ironhand the Warrior  Lv1  HP 100  MP 100  XP 7  Gold 1354  STR 770 DEX 600 AGI 550  weapon: none, armor: none
  2) Sehanine Monnbow the Warrior  Lv1  HP 100  MP 600  XP 8  Gold 2500  STR 770 DEX 500 AGI 880  weapon: none, armor: none
  3) Muamman Duathall the Warrior  Lv1  HP 100  MP 300  XP 6  Gold 2546  STR 990 DEX 750 AGI 550  weapon: none, armor: none
  4) Flandal Steelskin the Warrior  Lv1  HP 100  MP 200  XP 7  Gold 2500  STR 825 DEX 700 AGI 715  weapon: none, armor: none
  5) Undefeated Yoj the Warrior  Lv1  HP 100  MP 400  XP 7  Gold 2500  STR 880 DEX 700 AGI 440  weapon: none, armor: none
  6) Eunoia Cyn the Warrior  Lv1  HP 100  MP 400  XP 6  Gold 2500  STR 770 DEX 600 AGI 880  weapon: none, armor: none
  7) Rillifane Rallathil the Sorcerer  Lv1  HP 100  MP 1300  XP 9  Gold 2500  STR 750 DEX 550 AGI 495  weapon: none, armor: none
  8) Segojan Earthcaller the Sorcerer  Lv1  HP 100  MP 900  XP 5  Gold 2500  STR 800 DEX 715 AGI 550  weapon: none, armor: none
  9) Reign Havoc the Sorcerer  Lv1  HP 100  MP 800  XP 8  Gold 2500  STR 800 DEX 880 AGI 880  weapon: none, armor: none
  10) Reverie Ashels the Sorcerer  Lv1  HP 100  MP 900  XP 7  Gold 2500  STR 800 DEX 440 AGI 770  weapon: none, armor: none
  11) Kalabar the Sorcerer  Lv1  HP 100  MP 800  XP 6  Gold 2500  STR 850 DEX 660 AGI 440  weapon: none, armor: none
  12) Skye Soar the Sorcerer  Lv1  HP 100  MP 1000  XP 5  Gold 2500  STR 700 DEX 550 AGI 440  weapon: none, armor: none
  13) Parzival the Paladin  Lv1  HP 100  MP 300  XP 7  Gold 2500  STR 825 DEX 770 AGI 650  weapon: none, armor: none
  14) Sehanine Moonbow the Paladin  Lv1  HP 100  MP 300  XP 7  Gold 2500  STR 825 DEX 770 AGI 700  weapon: none, armor: none
  15) Skoraeus Stonebones the Paladin  Lv1  HP 100  MP 250  XP 4  Gold 2500  STR 715 DEX 385 AGI 600  weapon: none, armor: none
  16) Garl Glittergold the Paladin  Lv1  HP 100  MP 100  XP 5  Gold 2500  STR 660 DEX 440 AGI 500  weapon: none, armor: none
  17) Amaryllis Astra the Paladin  Lv1  HP 100  MP 500  XP 5  Gold 2500  STR 550 DEX 550 AGI 500  weapon: none, armor: none
  18) Caliber Heist the Paladin  Lv1  HP 100  MP 400  XP 8  Gold 2500  STR 440 DEX 440 AGI 400  weapon: none, armor: none
Pick hero #1: 2
Pick hero #2: 7
```


P = your party M = market X = blocked blank = wilderness (monsters!)
[W/A/S/D] move [I] info [E] equip/potions [M] market [H] help [Q] quit
> s
That way is blocked.

P = your party M = market X = blocked blank = wilderness (monsters!)
[W/A/S/D] move [I] info [E] equip/potions [M] market [H] help [Q] quit
> d
You arrived at a market. Press M to shop.

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| M |   | M | X |   |   | X |   |
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| M |   |   | M |   |   | X | M |
+---+---+---+---+---+---+---+---+
| M |   | X |   | X | M | M | X |
+---+---+---+---+---+---+---+---+
| X |   | M | X | X | M |   |   |
+---+---+---+---+---+---+---+---+
P = your party   M = market   X = blocked   blank = wilderness (monsters!)
[W/A/S/D] move  [I] info  [E] equip/potions  [M] market  [H] help  [Q] quit
> m
=== Welcome to the market! ===
Which hero is shopping?
  1) Sehanine Monnbow (Lv1, 2500 gold)
  2) Rillifane Rallathil (Lv1, 2500 gold)
  0) Leave the market
> 2
Rillifane Rallathil has 2500 gold.  1) Buy  2) Sell  0) Back
> 1
Buy which? (0 to cancel):
  1) Bow                    Lv2    300g  Weapon  damage 500, 2-handed, uses 10
  2) Scythe                 Lv6   1000g  Weapon  damage 1100, 2-handed, uses 10
  3) Axe                    Lv5    550g  Weapon  damage 850, 1-handed, uses 10
  4) Guardian Angel         Lv10  1000g  Armor   reduces damage by 1000, uses 10
  5) Healing Potion         Lv1    250g  Potion  +100 health
  6) Strength Potion        Lv1    200g  Potion  +75 strength
  7) Magic Potion           Lv2    350g  Potion  +100 mana
  8) Luck Elixir            Lv4    500g  Potion  +65 agility
  9) Mermaid Tears          Lv5    850g  Potion  +100 health/mana/strength/agility
  10) Ambrosia               Lv8   1000g  Potion  +150 health/mana/strength/dexterity/agility
  11) Snow Cannon            Lv2    500g  Ice spell  damage 650, mana 250
  12) Ice Blade              Lv1    250g  Ice spell  damage 450, mana 100
  13) Arctic Storm           Lv6    700g  Ice spell  damage 800, mana 300
  14) Flame Tornado          Lv4    700g  Fire spell  damage 850, mana 300
  15) Heat Wave              Lv2    450g  Fire spell  damage 600, mana 150
  16) Lava Comet             Lv7    800g  Fire spell  damage 1000, mana 550
  17) Hell Storm             Lv3    600g  Fire spell  damage 950, mana 600
  18) Thunder Blast          Lv4    750g  Lightning spell  damage 950, mana 400
  19) Spark Needles          Lv2    500g  Lightning spell  damage 600, mana 200
> 1
  You need to be level 2 to buy that.
Rillifane Rallathil has 2500 gold.  1) Buy  2) Sell  0) Back
```

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+---+---+---+---+---+---+---+---+
|   | M | X |   | M |   |   | M |
+---+---+---+---+---+---+---+---+
| X | M |   | M |   |   | X | M |
+---+---+---+---+---+---+---+---+
|   | P |   |   |   |   |   | M |
+---+---+---+---+---+---+---+---+
|   |   | M |   | X |   |   | M |
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| M |   | M | X |   |   | X |   |
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| M |   |   | M |   |   | X | M |
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| M |   | X |   | X | M | M | X |
+---+---+---+---+---+---+---+---+
| X |   | M | X | X | M |   |   |
+---+---+---+---+---+---+---+---+
P = your party   M = market   X = blocked   blank = wilderness (monsters!)
[W/A/S/D] move  [I] info  [E] equip/potions  [M] market  [H] help  [Q] quit
> s

*** A BATTLE BEGINS! ***
--- Heroes ---
  Sehanine Monnbow the Warrior  Lv1  HP 100  MP 600  weapon: none, armor: none
  Rillifane Rallathil the Sorcerer  Lv1  HP 100  MP 1300  weapon: none, armor: none
--- Monsters ---
  Andromalius the Spirit  Lv1  HP 100  DMG 37  DEF 150  DODGE 9%
  Aasterinian the Exoskeleton  Lv1  HP 100  DMG 20  DEF 138  DODGE 11%

--- Round 1: Heroes' turn ---

Sehanine Monnbow the Warrior  Lv1  HP 100  MP 600  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
>
```

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--- Round 1: Heroes' turn ---

Sehanine Monnbow the Warrior  Lv1  HP 100  MP 600  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Choose a target (0 to cancel):
  1) Andromalius the Spirit  Lv1  HP 100  DMG 37  DEF 150  DODGE 9%
  2) Aasterinian the Exoskeleton  Lv1  HP 100  DMG 20  DEF 138  DODGE 11%
> 1
Sehanine Monnbow attacked Andromalius for 31 damage!

Rillifane Rallathil the Sorcerer  Lv1  HP 100  MP 1300  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 4
  Rillifane Rallathil has no weapons or armor.

Rillifane Rallathil the Sorcerer  Lv1  HP 100  MP 1300  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Choose a target (0 to cancel):
  1) Andromalius the Spirit  Lv1  HP 69  DMG 37  DEF 150  DODGE 9%
  2) Aasterinian the Exoskeleton  Lv1  HP 100  DMG 20  DEF 138  DODGE 11%
> 2
Rillifane Rallathil attacked Aasterinian for 31 damage!
Andromalius attacked Rillifane Rallathil for 37 damage!
Rillifane Rallathil dodged Aasterinian's attack!

Surviving heroes recover some HP and MP.

--- Round 2: Heroes' turn ---

Sehanine Monnbow the Warrior  Lv1  HP 110  MP 660  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 4
  Sehanine Monnbow has no weapons or armor.

Sehanine Monnbow the Warrior  Lv1  HP 110  MP 660  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Choose a target (0 to cancel):
  1) Andromalius the Spirit  Lv1  HP 69  DMG 37  DEF 150  DODGE 9%
  2) Aasterinian the Exoskeleton  Lv1  HP 69  DMG 20  DEF 138  DODGE 11%
> 1
Sehanine Monnbow attacked Andromalius for 31 damage!

Rillifane Rallathil the Sorcerer  Lv1  HP 70  MP 1430  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 4
  Rillifane Rallathil has no weapons or armor.

Rillifane Rallathil the Sorcerer  Lv1  HP 70  MP 1430  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Choose a target (0 to cancel):
  1) Andromalius the Spirit  Lv1  HP 38  DMG 37  DEF 150  DODGE 9%
  2) Aasterinian the Exoskeleton  Lv1  HP 69  DMG 20  DEF 138  DODGE 11%
> 1
Andromalius dodged Rillifane Rallathil's attack!
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Choose a target (0 to cancel):
  1) Andromalius the Spirit  Lv1  HP 7  DMG 37  DEF 150  DODGE 9%
  2) Aasterinian the Exoskeleton  Lv1  HP 69  DMG 20  DEF 138  DODGE 11%
> 1
Rillifane Rallathil attacked Andromalius for 30 damage!
Andromalius was defeated!
Rillifane Rallathil dodged Aasterinian's attack!

Surviving heroes recover some HP and MP.

--- Round 4: Heroes' turn ---

Sehanine Monnbow the Warrior  Lv1  HP 109  MP 799  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Sehanine Monnbow attacked Aasterinian for 32 damage!

Rillifane Rallathil the Sorcerer  Lv1  HP 84  MP 1730  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Rillifane Rallathil attacked Aasterinian for 31 damage!
Sehanine Monnbow dodged Aasterinian's attack!

Surviving heroes recover some HP and MP.

--- Round 5: Heroes' turn ---

Sehanine Monnbow the Warrior  Lv1  HP 120  MP 878  weapon: none, armor: none
  1) Attack  2) Cast spell  3) Use potion  4) Equip  5) Show stats
> 1
Sehanine Monnbow attacked Aasterinian for 32 damage!
Aasterinian was defeated!

*** VICTORY! ***
Sehanine Monnbow earned 4 XP and 200 gold.  LEVEL UP! Now level 2!
Rillifane Rallathil earned 4 XP and 200 gold.  LEVEL UP! Now level 2!
Battle summary: 5 round(s), 248 damage dealt, 57 damage taken, 2 monster(s) defeated, 5 dodge(s).

+---+---+---+---+---+---+---+---+
|   | M | X |   | M |   |   | M |
+---+---+---+---+---+---+---+---+
| X | M |   | M |   |   | X | M |
+---+---+---+---+---+---+---+---+
|   |   |   |   |   |   |   | M |
+---+---+---+---+---+---+---+---+
|   | P | M |   | X |   |   | M |
+---+---+---+---+---+---+---+---+
| M |   | M | X |   |   | X |   |
+---+---+---+---+---+---+---+---+
| M |   |   | M |   |   | X | M |
+---+---+---+---+---+---+---+---+
| M |   | X |   | X | M | M | X |
+---+---+---+---+---+---+---+---+
| X |   | M | X | X | M |   |   |
+---+---+---+---+---+---+---+---+
P = your party   M = market   X = blocked   blank = wilderness (monsters!)
[W/A/S/D] move  [I] info  [E] equip/potions  [M] market  [H] help  [Q] quit
>
```